MIL-D-51031C
24 June 1971
SUPERSEDING
MIL-D-0051031B(MU)
15 December 1965 and
MIL-D-51031A
14 September 1962

MILITARY SPECIFICATION

DISPERSER, RIOT CONTROL AGENT,

PORTABLE, M3

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of disperser.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-17504 - Coating Compound, Acrylic, Clear.
MIL-T-50036 - Talc, Technical, T1.
MIL-G-51025 - Gun, Portable Riot Control Agent Disperser, M9.

FSC 1040

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL116-3-40 - Disperser, Riot Control Agent, Portable, M3.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite 49 CFR 171-179.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL116-3-40 and on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the disperser shall conform to the specifications and drawings listed on Drawing DL116-3-40 and subsidiary drawing.

3.2 Assembly. Prior to testing for leakage and functioning (3.3 and 3.4) each disperser shall be assembled as follows: The gun tubing connector shall be inserted into the hose fitting, and the aft end of the M9 gun (Drawing D116-4-16) shall be attached to the male hose fitting. The joints shall be tightened with a wrench. The quick disconnect coupling shall be connected to the tank assembly at the wearer's right-hand side.

3.3 Leakage. When the disperser is assembled in accordance with 3.2, the regulator shall reduce 2000 ± 50 pounds per square inch gage (psig) air to 40 ± 10 psig) when tested as specified in 4.4.4.1 (see 6.3).

3.4 Functioning. The disperser, when charged with 7 to 8 pounds of talc conforming to MIL-T-50036, shall discharge a minimum of 6 1/4 pounds of talc when tested as specified in 4.3.3 (see 6.3).

3.5 Safety disc. The low pressure safety disc shall burst between 125 and 175 psig when tested as specified in 4.4.4.2 (see 6.3).

3.6 Identification. Each disperser shall be identified with a serial number assigned in accordance with the contract or order (see 6.2).

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of dispersers shall be produced in accordance with this specification for examination and tests (see 4.3).

3.8 Workmanship. The disperser shall be free from foreign matter (oil, grease or dirt) and damage such as cracks, dents or deformation which could affect its intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five dispersers and five safety discs (one disc shall be taken at random from each square foot of material) shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination.

(a) Disperser. The sample dispersers shall be examined in accordance with all the requirements of this specification.

(b) Components. The sample components shall be examined and tested in accordance with the classification of defects. The AQL's do not apply.

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4.3.2.2 For tests.

(a) Nondestructive test. The sample dispersers shall be tested in accordance with 4.4.4.1 and 4.3.3.

(b) Destructive test. The sample safety discs shall be tested in accordance with 4.4.4.2.

4.3.3 Talc test. The test shall be conducted as follows: The assembled disperser shall be weighed prior to test, pressurized to 2000 ± 50 psig, charged with the specified amount of talc, functioned and reweighed after functioning. Functioning shall consist of firing the M9 gun in 5-second bursts until the disperser has exhausted all air pressure or talc. A sufficient interval shall be allowed between bursts to permit recovery of pressure in the fuel tanks.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4. Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. One safety disc shall be taken at random from each square foot of material.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample dispersers and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive test. The sample discs shall be tested in accordance with 4.4.4.2. The acceptance number is zero.

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4.4.3.3 For critical defects. Each item in the lot shall be tested for the critical characteristic listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Tank assembly, riot control agent disperser (Drawing E116-6-157).

Categories		Defects	Acceptance standards
Critical:			
1		Leakage	4.4.4.1
Major:	AQL	1.0 percent defective	
101		Component missing or incorrectly assembled	
102		Serial number missing	
103		Workmanship (3.8)	
Minor:	AQL	4.0 percent defective	
201		Protective finish missing	

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(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Components and repair parts not cleaned and dried as specified
102	Preservatives missing or incorrect
103	Packaging methods incorrect
104	Containers incorrect
105	Container closures incorrect
106	Components or repair parts missing
107	Container or package damaged (broken, crushed, punctured)
108	Strapping loose or missing
109	Marking missing, incorrect, or illegible
110	Preservative warning tag missing
111	Contents not complete (Drawing C116-3-46)
112	Contents arranged incorrectly

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. Tighten the agent container filler plugs by hand. Connect the air hoses, and connect a pressure gage between one air hose and one filler plug. Close the bleeder valve, and charge the air bottle with air to 2000 ± 50 psig. With the gun trigger in the forward position, open the high pressure valve. Read the pressure gage. Check all joints for leakage with a soap solution or other suitable leak-detecting solution. Open and close the bleeder valve. When open, the bleeder valve shall discharge air. Operate the gun in 5-second bursts to discharge air. During operation of the gun, the agent tank pressure may vary from the 40 ± 10 psig, but shall not exceed 90 psig.

4.4.4.2 Safety disc. Install the disc in a suitable adapter, and apply the specified pressure.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The disperser, components and repair parts shall be cleaned and dried in accordance with the applicable requirements of MIL-P-116, using cleaning method C-1 and any drying method listed therein.

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5.2 Packaging. Level shall be as specified (see 6.2). Unless otherwise specified, preservatives and packaging methods cited herein shall be in conformance to MIL-P-116.

5.2.1 Level A.

5.2.1.1 Unit preservation and packaging.

(a) Tank assembly. The threaded areas of the pipe and cap assembly and of the agent tanks shall be coated with type P-10, grade 2 or 3 preservative oil and installed in place. Air hoses shall be connected to the pipe and cap assemblies. A source of dry nitrogen shall be attached to the check valve on the pressure container, the pressure control valve and the shutoff valve shall be opened, and the system shall be purged. The shutoff valve shall be closed and the system charged with dry nitrogen to 5 psig. The pressure control valve shall then be closed, the dry nitrogen source disconnected from the check valve and the check valve cap secured. A tag conforming to UU-T-81, type B, class 1 or 2, shall be attached in a conspicuous place on the unit. The tag shall state that the system has been charged to 5 psig with dry nitrogen and that the shutoff and pressure control valves must be opened to discharge the system. Class 1 tags shall be waterproofed on both sides with acrylic coating conforming to MIL-C-17504.

(b) Gun, disperser, M9. The gun and components thereto shall be packaged in accordance with the level A requirements of MIL-G-51025.

(c) Hose assembly. The opening in each end of the hose shall be closed with plastic caps or plugs. The couplings shall then be cushion wrapped with grade A greaseproof barrier material conforming to MIL-B-121, securing the wrap in place with pressure-sensitive tape conforming to PPP-T-76.

(d) Gloves, rubber. Each pair of gloves shall be packaged in accordance with method IC-1 or IC-3.

(e) Tube, rubber. The tubing shall be packaged in accordance with the method cited in 5.2.1.1(d).

(f) Technical publications. Technical publications shall be packaged in accordance with method cited in 5.2.1.1(d). The two manuals shall be in one package and the DOA forms in one package.

(g) Packing, preformed (O-ring), gasket and lead washer. The repair parts shall be packaged together in accordance with method cited in 5.2.1.1(d).

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(h) Connector and head, safety. The connector shall be coated with P-7 preservative and like items packaged together using method IC-1.

(i) Valve check assembly. The valve shall be coated with P-7 preservative and individually packaged using method IC-1 or IC-4.

5.2.1.2 Intermediate packaging. Items unit-preserved and packaged as specified in 5.2.1.1(g), (h) and (i) shall be packaged together in a W5c, style RSC fiberboard box conforming to PPP-B-636 and internal cushioning used as required to make a tight package. All seams shall be taped including the manufacturer's joint with type II, class 2, minimum 2-inches wide tape conforming to PPP-T-76. The box dimensions, inside, shall approximate 6 1/2 by 5 by 3 1/4 inches, length, width and depth respectively.

5.2.2 Level C.

5.2.2.1 Unit preservation and packaging.

(a) Tank assembly. The tank assembly shall be prepared in accordance with 5.2.1.1(a) except the system is not required to be charged and the tag is not included.

(b) Gun, disperser, M9. The gun and components shall be packaged in accordance with level C packaging requirements of MIL-G-51025.

(c) Hose assembly. The hose assembly shall be prepared in accordance with 5.2.1.1(c).

(d) Rubber gloves. Each pair of gloves shall be maintained as a pair by uniting in any practicable manner.

(e) Tube, rubber. Each piece of tubing shall be coiled and secured in any practicable manner to facilitate handling, without damage.

(f) Technical publications. The documents shall be packaged together in a suitable bag or wrap.

(g) Packing, preformed (O-rings), gaskets and lead washers. The items shall be united, like items together, by tying, taping or bagging.

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(h) Connector and head, safety. The connector and safety head shall be wrapped in any practicable manner to keep items clean and to maintain identity.

(i) Valve check assembly. The valve shall be wrapped in any practicable manner to keep it clean and to maintain identity.

5.2.2.2 Intermediate packaging. Items unit-packaged in accordance with 5.2.2.1(g), (h) and (i) shall be intermediately packaged together in a fiberboard box conforming to type CF, style RSC, class domestic, grade 125, minimum, of PPP-B-636, having inside dimensions of approximately 6 1/2 by 5 by 3 1/4 inches, length, width and depth respectively. Appropriate internal cushioning material shall be included as required to restrict the items from free movement. The box shall be closed in accordance with closure requirements for domestic class boxes in the appendix to PPP-B-636.

5.3 Level A, packing. The disperser and components packaged as specified, shall be positioned within the chest as shown on Drawing E116-3-42. The chest shall be assembled as shown on Drawings E116-3-43 and E116-3-44. The chest shall be closed and strapped as shown on Drawing E116-3-42.

5.4 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein.

5.4.1 Package. Unit and intermediate packages shall be marked with item identification markings, including, quantity per package, in accordance with MIL-STD-129.

5.4.2 Packing. Containers shall be marked in accordance with Drawing D116-3-53.

5.5 Palletization. When specified in the contract or order (mandatory for level A shipments), the packed dispersers shall be palletized and the unit load marked in accordance with Drawing DL116-3-52.

6. NOTES

6.1 Intended use. This item is intended for use in suppressing riots or dispersing mobs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Serial numbers for each disperser.

(c) Preproduction.

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(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

(d) Palletization when required.

6.3 Caution. Due to the hazards inherent in testing with high pressures, suppliers are hereby notified that every precaution possible should be taken for the safety and protection of personnel engaged in testing.

6.4 Component specifications. The following specifications and SQAP's are required for the completion of this item:

MIL-R-11534	-	Regulator, Pressure.
MIL-H-51059	-	Hose, Fuel.
MIL-C-51234	-	Coupling Assembly, Quick-Disconnect.
SQAP 116-6-148	-	Tube Assembly.
SQAP 116-6-155	-	Adapter, Safety Valve.
SQAP 116-6-158	-	Hose Assembly.
SQAP 116-6-161	-	Valve, Shutoff, Modification of.
SQAP 81-1-439	-	Valve, Pressure Tank.
SQAP 81-1-470	-	Shaft, Flexible, Valve Assembly.
SQAP 81-1-483	-	Tank, Fuel.
SQAP 81-1-504	-	Frame Carrier.
SQAP 81-1-879	-	Tank and Valve Pressure Assembly.
SQAP 81-1-884	-	Coupling Half, Quick-Disconnect.
SQAP 81-1-1796	-	Valve, Safety.
SQAP 81-1-1797	-	Adapter, Safety Valve.
SQAP 81-1-1799	-	Sping, Helical Compression.
SQAP 81-1-2146	-	Valve Assembly, Check.

6.5 Pressure regulator. The following method of setting the pressure regulator has been found satisfactory: A 100 ± 10 psig air source, a valve, the regulator and an outlet pressure gage, connected in that order. The regulator should be adjusted to 40 + 5 psig following the method in TM3-1040-214-12 paragraph 51e-(4).

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Custodian:

Army - MU
Navy - MC
Air Force - 84

Review interest:

Navy - MC
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Preparing activity:

Army - MU(EA)

Project No. 1040-0086

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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